Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSDE711]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

PATTERN RECOGNITION

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Illustrate the design principles of pattern recognition system with an example.	08
	b.	Calculating conditional probability of rain whether records show that barometric pressure occurred on 160 of the 200 days in a data set and it rained on 20 of the 160 days.	06
2.	a.	Why feature Extraction is necessary? Explain the concept of feature extraction in pattern recognition system with examples.	08
	b.	Describe Bayes decision rule. Explain how it can be used for two class classification.	06
3.	a.	Explain the Principle Component Analysis for dimension reduction.	08
	b.	Describe Statistical Patten Recognition with example.	06
		OR	
4.	a.	With expressions describe Any path method and Best Path Method of recognition in the Hidden Markov Models.	08
	b.	Discuss the Fisher Linear discriminant analysis with suitable example.	06
5.	a.	With suitable examples, Explain univariate and multivariate normal density functions.	08
	b.	Explain the Density Estimation with suitable example	06
		OR	
6.	a.	Consider the set of two dimensional patterns denoted as **(x-coordinate, y-coordinate, class belongs to)** with two classes as **class 1 and class 2.** (1, 1, 1), (1, 2, 1), (1 ,3, 1), (2, 1, 1), (2, 2, 1), (2, 3, 1), (2, 3.5, 1), (3.5, 2, 2), (5, 4, 2), (5, 5, 2), (6, 3, 2), (6, 4, 2), (4.5, 2, 2), and (6, 5, 2). **If P is at (3.8, 3.1), Find the class of P using kNN algorithm where k=3.**	08
	b.	Why hierarchical clustering is better than K-Means Clustering Algorithm	06
7.	a.	Describe the working and application of agglomerative hierarchical clustering,	08
	b.	Explain the advantages and disadvantages of hierarchical clustering.	06
		OR	
8.	a.	Examine the Fuzzy C Mean clustering with appropriate example	08
	b.	Explain the Unsupervised Learning and Clustering.	06